AI/ML INTERNSHIP · DEVELOPERSHUB CORP

Exploring the *Iris Dataset*

Exploratory Data Analysis · Python · Visualization

01
TASK ONE

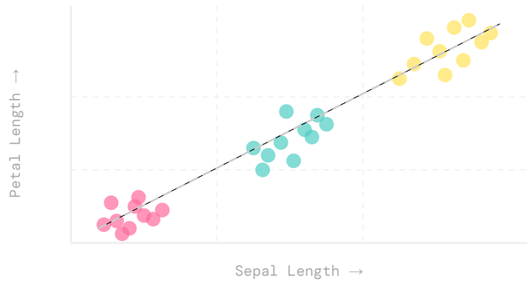
by Abdul Samad

Iris Setosa · Iris Versicolor · Iris Virginica

150 samples · 4 features · 3 classes

PLOT 01 — SCATTER

Feature Relationships

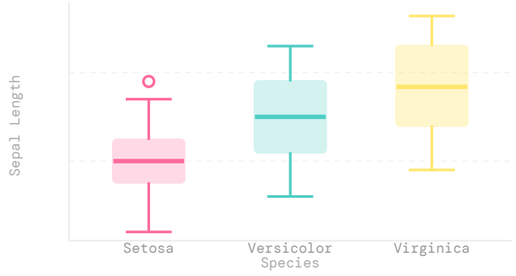


PLOT 02 — HISTOGRAM

Value Distributions

PLOT 03 — BOX PLOT

Outlier Detection



DATASET OVERVIEW

Iris Dataset

150 SAMPLES	**4** FEATURES
3 CLASSES	**0** MISSING VALS

SOURCE

```
seaborn.load_dataset('iris')
```

KEY FINDINGS

What the Data Revealed

Setosa is Linearly Separable 01

Clearly distinct from other species based on petal dimensions alone — ideal for simple classifiers.

Bimodal Petal Distributions 02

Petal length and width show two distinct peaks in histograms — a direct signature of class groupings.

Versicolor & Virginica Overlap 03

These two species share similar feature ranges, making them harder to distinguish without ML models.

TOOLS USED

Libraries

Pandas
Data loading & inspection

Seaborn
Statistical visualization

Matplotlib
Plot customization

SKILLS PRACTICED

What I Learned

- Data Loading
- EDA
- Descriptive Stats
- Scatter Plots
- Histograms
- Box Plots
- Outlier Detection
- Data Cleaning
- Pattern Recognition

WORKFLOW

Steps

1. **Load** — pandas read dataset
2. **Inspect** — .head() .info() .describe()
3. **Scatter** — Feature relationships
4. **Histogram** — Value distributions
5. **Box Plot** — Outlier detection
6. **Insights** — Draw conclusions

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